

Noise-Robust Bird Vocalization Classification for Low-Resource Scenarios

1) Problem Statement

Core Machine Learning Task

Multi-class audio classification: identify wild bird species from field-recorded audio with long-tailed class distribution and complex environmental background noise.

Literature Review & Existing Research Gaps

We systematically reviewed BirdNET official publications, BirdCLEF competition papers and arXiv bioacoustic articles:

1. Mainstream off-the-shelf tools: BirdNET (Cornell Lab + Chemnitz University, <https://birdnet.cornell.edu/>) and Google Perch both rely on single fixed ResNet CNN architectures trained on large-scale global bird datasets.
2. Critical limitations verified in existing literature:
 1. BirdNET only passively processes unquantified, random natural field noise, lacking standardized controllable noise testing pipelines;
 2. Severe performance degradation occurs on low-resource rare bird species with insufficient training samples;
 3. No unified open-source framework simultaneously compares traditional tabular ML, lightweight image model and raw-audio transfer learning on identical bird audio datasets;
 4. Existing papers either test single deep learning architectures or only compare multiple deep learning variants; few integrate gradient-boosted tree models for horizontal benchmarking.
2. Research gap summary: No reproducible controlled-variable experiment exists to separate clean baseline audio and programmatically generated gradient noise for systematic anti-robustness comparison, which is the core novelty of our project.

Standardized Controlled-Variable Experimental Paradigm (Academic Norm)

We design two separated experimental groups for fully controllable, repeatable testing:

1. Baseline Control Group (simulate ideal laboratory zero-interference recording environment): From the full 12,400 field audio dataset, we extract high-SNR

(≥ 15 dB) clean recordings with minimal natural noise and single-species vocalization only. This subset acts as standardized baseline to measure native classification accuracy of each model under ideal low-disturbance conditions.

2. Artificial Noise Test Group (quantifiable robustness evaluation): Based on the above clean baseline samples, we programmatically overlay gradient-controllable noise (Gaussian white noise, low-frequency wind noise, intermittent pulse noise) at adjustable SNR tiers (5 dB / 0 dB / -5 dB). All noise intensity is fully quantified to complete objective anti-noise performance comparison, a design absent in BirdNET's native workflow.

Proposed Three Independent ML Solutions & Novelty Compliance

We implement three fundamentally distinct algorithms with different input modalities on one unified dataset, fully complying with the rule of "no ready-made complete end-to-end solution online":

1. LightGBM (tabular handcrafted audio statistics, traditional machine learning)
 2. FastAI lightweight image model (mel-spectrogram image input, computer vision deep learning)
 3. YAMNet transfer learning (raw audio waveform input, pre-trained audio foundation model)
- No identical three-model comparative pipeline combined with controllable artificial noise augmentation can be found in BirdNET official code, Kaggle notebooks or peer-reviewed bioacoustic papers.

Practical Value

Provide ecologists with a low-cost, reproducible benchmark pipeline to select optimal models for both abundant common birds and scarce low-resource endangered species, reducing manual audio screening workload for biodiversity monitoring.

2) Description of Data Set

Dataset Source & URL

Main dataset: Google Perch bird vocalization classifier dataset on Kaggle (<https://www.kaggle.com/models/google/bird-vocalization-classifier>).

Basic Dataset Profile

- Data format: 10-second mono WAV raw audio + complete metadata (bird species tag, background noise grade, SNR value, multi-species overlap label)
- Total scale: 12,400 field wild audio clips covering 86 distinct bird species
- Core inherent challenges:

1. Severe long-tailed imbalance: Common species exceed 800 samples per category; low-resource rare species have fewer than 50 recordings each;
2. 62% clips contain mixed wind, rain, insect and frog natural background noise;
3. Partial samples hold overlapping vocalizations from multiple bird species, leading to ambiguous labels.

Representative Stratified Subset Selection (Eliminate Sampling Bias)

Due to limited team computing resources and weekend-only working schedule, we adopt dual-layer proportional stratified sampling before filtering clean baseline samples to guarantee statistical representativeness:

1. Stratification dimensions: Bird species label + metadata SNR tier (High SNR ≥ 15 dB / Medium 0–15 dB / Heavy < 0 dB);
2. Hard constraint: Every bird species retains a minimum of 5 samples to avoid complete loss of low-resource rare birds;
3. Sample size: Extract a balanced representative subset of roughly 2,000 clips from the full 12,400-record pool;
4. Representativeness verification: Compare species proportion and three-tier noise distribution between full dataset and 2,000-subset; control proportion deviation within $\pm 3\%$ to prove the subset can reliably reflect the original data distribution.
5. Post-sample high-SNR filtering: Inside the balanced 2,000-subset, we further screen high-SNR clean single-species recordings as the lab-simulation baseline control group; three models will use this unified filtered subset for training and evaluation to guarantee fair comparison.

Self-Designed Preprocessing, Two-Stage Training Augmentation, and Test-Time Noise Injection (Not Supported by BirdNET)

1. Feature extraction branch 1 (for LightGBM): Compute handcrafted tabular audio features: spectral centroid, bandwidth, rolloff, MFCC coefficients, zero-crossing rate;
2. Feature extraction branch 2 (for FastAI): Convert raw waveforms to standardized mel-spectrogram 2D images;
3. Exclusive double-layer augmentation pipeline:
 - Waveform-level augmentation: White noise injection, time masking, pitch shifting, volume scaling;

- Spectrogram-level exclusive augmentation for FastAI: Random brightness adjustment, partial occlusion, minor zoom transform;
- 4. Controlled artificial noise generation module: Overlay three types of adjustable-intensity noise onto clean baseline clips to build standardized noisy test sets for robustness quantification.

3) Implementation Plan (4 Weekend Schedule, Weekday Automatic Training)

Weekend 1: Literature Review & Stratified Data Sampling

Saturday: Summarize limitations of BirdNET, Google Perch and existing bioacoustic papers for the Related Work chapter; download Kaggle dataset and read metadata CSV. Sunday: Code dual-layer stratified sampling logic, extract 2,000 representative balanced subset and verify distribution consistency with full dataset; filter high-SNR clean baseline audio inside the subset. Weekday automatic task: Run preliminary data cleaning and imbalance resampling scripts in the background.

Weekend 2: Unified Preprocessing & Noise Augmentation Development

Saturday: Write shared audio processing code to extract tabular features and generate mel-spectrogram images; implement self-built two-stage training augmentation and test-time noise injection module. Sunday: Split the balanced subset into train / validation / test partitions following a 70/15/15 ratio. Weekday automatic task: Launch LightGBM baseline model training in idle computer time.

Weekend 3: FastAI & YAMNet Transfer Learning Construction

Saturday: Independently build, configure and train FastAI lightweight image model on mel-spectrogram images; apply lr_find, early stopping and dropout regularization for hyperparameter optimization. Sunday: Load pre-trained YAMNet weights, implement layer-wise fine-tuning strategies for raw waveform input, complete model checkpoint saving.

Weekend 4: Unified Model Evaluation & Final Report Compilation

Saturday: Build universal evaluation script to calculate test accuracy, training runtime, single-sample inference latency and GPU memory consumption for all three algorithms; generate confusion matrices and comparative performance charts. Sunday: Analyze strengths, weaknesses and anti-noise robustness of each model; integrate all team outputs into the full 7-section final report.

Three Independent ML Algorithm Implementation Details

Solution 1: LightGBM Gradient Boosting Tree (Traditional Tabular ML, Jianan Zhang)

Input: Handcrafted normalized audio statistical features (no image conversion required). Workflow: Grid search for hyperparameters (tree number, max depth, leaf count, class weight for imbalance); evaluate via precision, recall, F1-score and overall test accuracy. Advantages: Ultra-low hardware demand, fast training & inference, interpretable feature importance ranking. Disadvantages: Unable to capture spatial texture features of mel-spectrograms, weak classification performance on heavily overlapping multi-bird audio.

Solution 2: FastAI Lightweight Mel-Spectrogram Image Model (Image Deep Learning, Jincheng Chen)

Input: Augmented standardized mel-spectrogram 2D tensors. Workflow: Build a compact image-classification backbone with dropout regularization via FastAI high-level API; tune batch size, learning rate and image-model hyperparameters through ablation experiments; save optimal checkpoint via early stopping. Same audio-to-image logic as BirdNET but equipped with two-stage augmentation absent in BirdNET's native pipeline. Advantages: Optimized image feature extraction workflow with built-in automation tools, lighter inference cost than large BirdNET ResNet. Disadvantages: Prone to overfitting on low-resource rare bird species with limited samples.

Solution 3: Fine-Tuned YAMNet Transfer Learning (Raw Audio Transfer Model, Wenjuan Huang)

Input: Unconverted raw audio waveform, direct audio-domain feature extraction. Workflow: Freeze partial pre-trained YAMNet backbone layers, unfreeze top convolution blocks for bird dataset fine-tuning; replace original classification head with a custom dense layer matching our 86 species; test multiple differential learning rate strategies for frozen/unfrozen layers. Advantages: Best anti-noise generalization performance on low-SNR noisy field recordings among the three models. Disadvantages: Consumes the maximum GPU memory and training computational cost.

Final Project Deliverables

1. One-page project proposal PDF (this submission document)

2. Complete final report covering 7 required chapters: Introduction, Related Work, Dataset Description, Algorithm Implementation, Parameter Tuning & Result Comparison, Future Research, Conclusion & References
3. Separate runnable Python scripts for each model + shared audio preprocessing code (tabular feature extraction + mel-spectrogram generation) and the controlled artificial noise injection module; plus the FastAI-specific two-stage augmentation module as a separate script.
4. Visualized comparison outputs: Confusion matrices, accuracy bar charts, computational cost curves, noise accuracy decay graphs

4) Team Members & Equal Task Allocation (One Full Pipeline Per Person)

Team size: 3 members, each independently responsible for one complete algorithm pipeline and corresponding report writing work.

Jianan Zhang – Literature Research, Data Pipeline & LightGBM Model

1. Complete full literature survey on BirdNET and existing bird classification research, draft the Related Work section;
2. Download Kaggle dataset, implement dual-layer stratified sampling, subset representativeness verification and high-SNR baseline filtering code;
3. Develop full LightGBM pipeline including handcrafted feature extraction, grid search hyperparameter tuning and performance metric logging;
4. Write Dataset Description and Traditional Machine Learning sections of the final report.

Jincheng Chen – FastAI Mel-Spectrogram Image Model Full Pipeline

1. Implement audio-to-mel-spectrogram conversion and FastAI image dataset loading pipeline;
2. Configure lightweight image model backbone, built-in image augmentation, learning rate search and training early stopping callbacks;
3. Record training loss/accuracy curves, save optimal model checkpoints and analyze the model's performance pros & cons;
4. Complete FastAI model design, training and parameter tuning analysis in the report.

Wenjuan Huang – YAMNet Transfer Learning, Unified Evaluation & Final Report Integration

1. Independently implement multi-strategy fine-tuning for pre-trained YAMNet raw audio model;
2. Develop universal evaluation framework to measure accuracy, inference speed and memory overhead of all three models; generate all comparison visualization charts;
3. Compare three models' noise robustness and low-resource species generalization ability; write Conclusion and Future Research chapters;
4. Integrate all team writing content, organize reference list and export the complete final PDF report.

Supplementary Compliance & Literature Remarks

1. Novelty Guarantee: The combination of three heterogeneous ML paradigms + dual-layer noise augmentation + fully controllable SNR gradient noise experiment cannot be found in BirdNET open-source code, Kaggle notebooks or peer-reviewed bioacoustic papers, satisfying the instructor's rule of prohibiting ready-made complete off-the-shelf solutions.
2. Algorithm Compliance: Three completely distinct machine learning categories are implemented for comprehensive horizontal comparison as required: tabular gradient boosting tree, image lightweight deep learning model, raw audio pre-trained transfer learning.
3. Cited Official Reference Links
 - BirdNET Official Homepage: <https://birdnet.cornell.edu/>
 - BirdNET-Analyzer GitHub Repository:
<https://github.com/kahst/BirdNET-Analyzer>
 - BirdNET-Analyzer Documentation:
<https://birdnet-team.github.io/BirdNET-Analyzer/>
 - Kaggle Google Perch Bird Vocalization Dataset:
<https://www.kaggle.com/models/google/bird-vocalization-classifier>